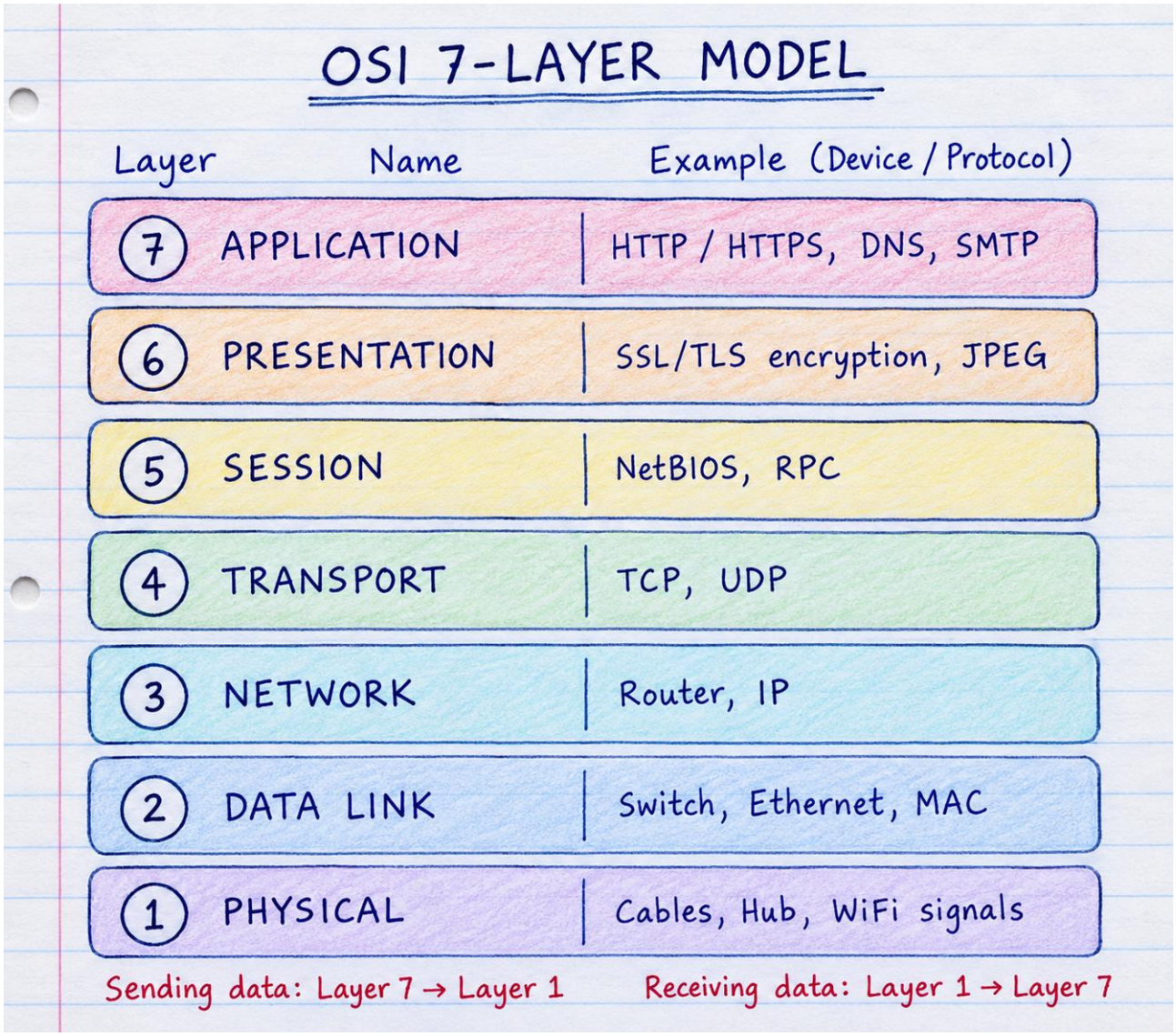
OSI Model Assignment

Q1. Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

Answer:



Q2. Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.

Answer:

a) Encrypting data before sending: **Presentation Layer (6)**

b) Routing packets across networks: **Network Layer (3)**

c) Converting data to electrical signals: **Physical Layer (1)**

d) Opening a web page in your browser: **Application Layer (7)**

e) Detecting and correcting errors in frames: **Data Link Layer (2)**

Q3. Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

Answer:

App chosen: Instagram (when I upload a photo)

- **Application Layer (7):** The Instagram app creates the upload request and talks to Instagram's servers using HTTPS.
- **Presentation Layer (6):** The photo is compressed (JPEG) and encrypted with SSL/TLS so no one can read it on the way.
- **Transport Layer (4):** TCP breaks the photo into small segments, numbers them and makes sure all of them reach the server (lost ones are resent).
- **Network Layer (3):** IP adds the source and destination IP addresses, and routers send the packets towards Instagram's server.
- **Data Link Layer (2):** WiFi/Ethernet puts the data into frames and uses MAC addresses to move it from my phone to the router.
- **Physical Layer (1):** The data is sent as radio waves (WiFi) or electrical/light signals through cables.

When I receive data (like a reel), the same steps happen in reverse order, from Layer 1 up to Layer 7.

Q4. Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning

Answer:

Layers working fine: Ping works at the Network Layer (it uses IP/ICMP). A successful ping means the Physical (1), Data Link (2) and Network (3) layers are working, so my cable/WiFi, router and IP connection are all fine.

Layers that might be causing the problem: Since the lower layers are fine, the problem is in the higher layers (4 to 7):

- **Application Layer (7) - most likely:** DNS may not be converting the website name into its IP address, or the browser/HTTP/HTTPS request is failing, or the website's web service is down.
- **Transport Layer (4):** A firewall may be blocking the web ports (80 for HTTP, 443 for HTTPS), even though ping is allowed.
- **Presentation Layer (6):** An SSL/TLS certificate problem can stop the secure website from opening.

Conclusion: The problem is most likely at the Application Layer (DNS or HTTP) or the Transport Layer (blocked port), not in the physical connection. Possible fixes: check DNS (for example, try 8.8.8.8), clear browser cache, and check firewall settings.